

Advanced Data Management

2022/2023

Practical assignment #3bis: Modeling in MongoDB and Cassandra

Aim

The aim of this optional assignment is to learn: (i) how to model aggregates in MongoDB and Cassandra; (ii) how to change consistency levels in Cassandra, according to the quorum formula for leader-less systems (see the slides for details). Please refer to the previous assignments for information about the cluster and modalities for accessing MongoDB and Cassandra.

Exercise 1: Query-driven design in MongoDB

1. For at least one request presented in the Practical Assignment #2 for which the data organization was in your opinion non-optimal, propose a new data organization (different aggregate and/or shard key and/or indexes), obtained by modifying the original ones, in which the request can be executed in a better way. Justify your answer.
2. Choose one query among those identified in point (1) above. Focus on a query which execution can be improved by indexes.

Create now your own database, indexing the collection as you deem appropriate for executing the query. To do so, take a look at this documentation page:

<https://docs.mongodb.com/manual/reference/method/db.collection.createIndex>.

3. Insert some rows inside the created table.
4. Show through `explain()` the differences in executing the query.
5. Present some other queries over the new created collection that can take advantage of indexes. Create such indexes and show how the queries are then executed.

Exercise 2: Query-driven design in Cassandra

1. Select one non admitted request from the Practical Assignment #3 (call it Q) and propose a new Cassandra table T , obtained by modifying the original one, for which the request is admitted. Justify your answer.
2. Create T in your own keyspace (named `ks_<userId>`).
3. Insert some rows inside T . Try to insert two rows with the same primary key. What does it happen?

4. Show that Q is now admitted when executed over the new table.
5. Present one query Q' over T that can be executed only by creating some indexes. Create such indexes and show that Q' can now be executed.

Exercise 3: Consistency levels in Cassandra

1. From `cqlsh` console, use `consistency` command to show the current consistency level.
2. Look at the slides and modify the consistency level to `TWO` and execute a select operation and a write operation over the table created in Exercise 2.
3. Modify the consistency level to `THREE` and execute a select operation and a write operation over the table created in Exercise 2.
4. Modify the consistency level to `QUORUM` and execute a select operation and a write operation over the table created in Exercise 2.
5. Modify the consistency level to `ALL` and execute a select operation and a write operation over the table created in Exercise 2.
6. Discuss in your report what you have done with the commands above.